# POINTS OF A LOCALIZED SITE LYING OVER A GIVEN POINT

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

*Allowed standard prerequisites (editorial).*

- The point of a localized site attached to an element. Same chapter, Lemma lemma-point-localize, recorded as the context of this card. Supplies the construction whose fibres are described by the correspondence.
- Points of a site versus points of its topos. Same chapter, Lemma lemma-point-site-topos. Identifies the values of the functor attached to a point with the pullback of representable sheaves, which is what is used to compute the diagrams.

*Source reference.* Original lemma 00YA: sources/stacks-sites/pinned-original-source.tex, source lines 8981–8981. Navigation pointer invoked at proof line 8981; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 00XM: sources/stacks-sites/pinned-original-source.tex, source lines 9006–9006. Navigation pointer invoked at proof line 9006; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 00WJ: sources/stacks-sites/pinned-original-source.tex, source lines 9009–9009. Navigation pointer invoked at proof line 9009; the counted

*Date*: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *Let $\mathcal{C}$ be a site. Let $p$ be a point of $\mathcal{C}$ given by $u : \mathcal{C} \to Sets$. Let $U$ be an object of $\mathcal{C}$ and let $x \in u(U)$. The functor*

$$v : \mathcal{C}/U \longrightarrow Sets, \quad (\varphi : V \to U) \longmapsto \{y \in u(V) \mid u(\varphi)(y) = x\}$$

*defines a point $q$ of the site $\mathcal{C}/U$ such that the diagram*

$$\begin{array}{ccc} & & Sh(pt) \\ & \overset{q}{\swarrow} & \downarrow p \\ Sh(\mathcal{C}/U) & \xrightarrow{j_U} & Sh(\mathcal{C}) \end{array}$$

*commutes. In other words $\mathcal{F}_p = (j_U^{-1}\mathcal{F})_q$ for any sheaf on $\mathcal{C}$.*

**Lemma 0.2.** *Let $\mathcal{C}$, $p$, $u$, $U$ be as in Lemma 0.1. The construction of Lemma 0.1 gives a one to one correspondence between points $q$ of $\mathcal{C}/U$ lying over $p$ and elements $x$ of $u(U)$.*

*Proof.* Let $q$ be a point of $\mathcal{C}/U$ given by the functor $v : \mathcal{C}/U \to Sets$ such that $j_U \circ q = p$ as morphisms of topoi. Recall that $u(V) = p^{-1}(h_V^\#)$ for any object $V$ of $\mathcal{C}$, see Lemma 00YA. Similarly $v(V/U) = q^{-1}(h_{V/U}^\#)$ for any object $V/U$ of $\mathcal{C}/U$. Consider the following two diagrams

$$\begin{array}{ccc} \mathrm{Mor}_{\mathcal{C}/U}(W/U, V/U) & \longrightarrow & \mathrm{Mor}_{\mathcal{C}}(W, V) \\ \downarrow & & \downarrow \\ \mathrm{Mor}_{\mathcal{C}/U}(W/U, U/U) & \longrightarrow & \mathrm{Mor}_{\mathcal{C}}(W, U) \end{array} \qquad \begin{array}{ccc} h_{V/U}^\# & \longrightarrow & j_U^{-1}(h_V^\#) \\ \downarrow & & \downarrow \\ h_{U/U}^\# & \longrightarrow & j_U^{-1}(h_U^\#) \end{array}$$

The right hand diagram is the sheafification of the diagram of presheaves on $\mathcal{C}/U$ which maps $W/U$ to the left hand diagram of sets. (There is a small technical point to make here, namely, that we have $(j_U^{-1}h_V)^\# = j_U^{-1}(h_V^\#)$ and similarly for $h_U$, see Lemma 00XM.) Note that the left hand diagram of sets is cartesian. Since sheafification is exact (Lemma 00WJ) we conclude that the right hand diagram is cartesian.

Apply the exact functor $q^{-1}$ to the right hand diagram to get a cartesian diagram

$$\begin{array}{ccc} v(V/U) & \longrightarrow & u(V) \\ \downarrow & & \downarrow \\ v(U/U) & \longrightarrow & u(U) \end{array}$$

of sets. Here we have used that $q^{-1} \circ j^{-1} = p^{-1}$. Since $U/U$ is a final object of $\mathcal{C}/U$ we see that $v(U/U)$ is a singleton. Hence the image of $v(U/U)$ in $u(U)$ is an element $x$, and the top horizontal map gives a bijection $v(V/U) \to \{y \in u(V) \mid y \mapsto x \text{ in } u(U)\}$ as desired. □